

sors via the core logic in the microcontroller / microprocessor board and thus helps in automating various parts of a home.

- **Human interface software**

- This software acts as an interface between humans and the microcontroller / microprocessor board.
- A home automation app running on smartphone is a good example.

For Arduino Users:

There is a fixed procedure that needs to be followed to upload the code of an Arduino project to the Arduino board. Please follow the link <http://www.ladyada.net/learn/arduino/lesson1.html> to get detailed instructions for the same.

DIY projects

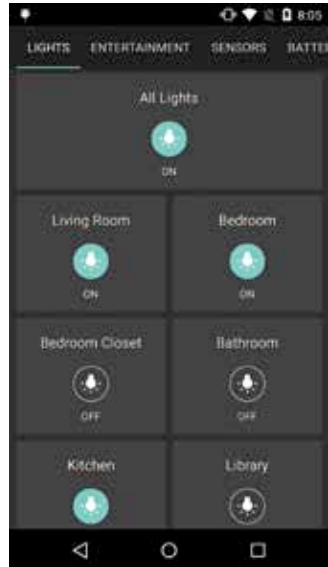
We will now have a look at some home automation projects that you can implement easily. Providing detailed instructions for projects is beyond the scope of this book. So, we have compiled a list of some interesting projects that can be a good starting point.

Water level indicator and controller using Arduino

This DIY uses ultrasonic distance sensor. Ultrasonic sensors send an ultrasonic pulse and measure the time difference between sending and receiving the pulse back. This time difference is used to measure distance of an object from ultrasonic sensor.

Components needed:

1. Arduino Uno
2. Ultrasonic sensor Module
3. 16x2 LCD
4. Relay 6 Volt
5. ULN2003 Relay circuit driver



Android app for human interaction